

The Degenerate Limit $c \rightarrow 0$ of the Shifted Krein Laplacian (Quotient Left-Definite Space, Structural Stability of the Krein–Sobolev System, and a Correction to the “Finite-Limit” Heuristic)

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Abstract

We analyze the limit $c \rightarrow 0$ of the first left-definite space $(H^1[-1, 1], (\cdot, \cdot)_{1,c})$ associated with the shifted Krein Laplacian $K_c f = -f'' + cf$ on $[-1, 1]$. The limit inner product $(\cdot, \cdot)_{1,0}$ is only a pseudo inner product whose radical is exactly $W = \text{span}\{1, x\}$ (the null space of K_0); we prove that the quotient H^1/W is a Hilbert space isometrically isomorphic to $L_0^2(-1, 1) = \{g \in L^2 : \int_{-1}^1 g = 0\}$, and that the images of the polynomials $\{S_n = P_n - P_{n-2} : n \geq 2\}$ span the image of all polynomials exactly, hence any Gram–Schmidt orthonormalization of them is a complete orthonormal system of the quotient. For the Krein–Sobolev system $\{K_n^{(c)}\}$ we prove: $K_0 = 1$, $K_1 = x$ have squared norms $2c$, $2c/3$ (vanish linearly); $K_2 = P_2$, $K_3 = P_3$ are c -independent, and their squared norms tend to 6, 10, respectively. For each fixed $n \geq 4$ the coefficients a_n and the squared norms $\|K_n^{(c)}\|_{1,c}^2 = 2c a_n a_{n+2}/(2n+1)$ diverge (e.g. $\|K_4^{(c)}\|_{1,c}^2 \sim 3150/c^2$), while the *unit-normalized* system $\hat{u}_n^{(c)} = K_n^{(c)}/\|K_n^{(c)}\|_{1,c}$ has, for every fixed $n \geq 2$, a limit in the ordinary Sobolev H^1 norm. With $Q_n = S_n/\sqrt{2(2n-1)}$, the representatives converge to $P_2/\sqrt{6} = Q_2 + 1/\sqrt{6}$ for $n = 2$, to $P_3/\sqrt{10} = Q_3 + x/\sqrt{10}$ for $n = 3$, and to Q_n for $n \geq 4$. Thus $[\hat{u}_n^{(c)}] \rightarrow [Q_n]$ in H^1/W for every fixed $n \geq 2$, and these limiting classes form a complete orthonormal system. This corrects both the low-mode representative limits and the earlier heuristic that “all norms for $n \geq 2$ have finite limits” (which holds only for $n = 2, 3$). *Proof status:* explicit Gram identities, parity induction with a controlled remainder, and finite polynomial expansions give full analytical proofs of these fixed-index assertions (Lemma 3.2, Theorems 4.2, 4.3). Finite checks are supplementary, and this author revision does not itself constitute independent acceptance. No convergence estimate uniform in n , or limit with n varying with c , is asserted.

1 Setting

Let $H^1 = H^1[-1, 1]$ over $\mathbb{C}$, and let Π denote the polynomials. Inner products are linear in the first argument; the real case follows by restriction. For $c \geq 0$ consider the sesquilinear form [1, eq. (2)]

$$(f, g)_{1,c} := -\frac{1}{2}(f(1) - f(-1))\overline{(g(1) - g(-1))} + \int_{-1}^1 (f'(x)\overline{g'(x)} + c f(x)\overline{g(x)}) dx, \quad (1)$$

For $c > 0$ this is the inner product of the first left-definite space of the shifted Krein Laplacian [1, eqs. (3)–(4)]

$$K_c f := -f'' + cf, \quad D(K_c) := \{f \in \Delta : f'(1) = f'(-1) = \frac{f(1) - f(-1)}{2}\}, \\ \Delta := \{f : f, f' \in AC[-1, 1], f'' \in L^2\}.$$

Integration by parts gives

$$(f, g)_{1,c} = (K_c f, g)_{L^2} \quad (f \in D(K_c), g \in H^1).$$

For $c > 0$, K_c is self-adjoint and $K_c \geq cI$. Its form domain and form satisfy [1, Theorem 1 and the following paragraph]

$$D(K_c^{1/2}) = H^1, \quad (f, g)_{1,c} = (K_c^{1/2} f, K_c^{1/2} g)_{L^2} \quad (f, g \in H^1). \quad (2)$$

Thus the square-root identity applies to all of H^1 , without imposing the operator boundary conditions on f or g . The spectrum is $\sigma(K_c) = \sigma(K_0) + c$ where [1, eqs. (7)–(8)]

$$\sigma(K_0) = \{0\} \cup \{(n\pi)^2 : n \geq 1\} \cup \{\mu_n^2 : n \geq 1\},$$

and μ_n is the unique root of $\tan \mu = \mu$ in $(n\pi, (n + \frac{1}{2})\pi)$, $n \geq 1$. The corresponding complete L^2 eigenfunction system is

$$\{1, x\} \cup \{\cos n\pi x : n \geq 1\} \cup \{\sin \mu_n x : n \geq 1\}.$$

The null space of K_0 is $W := \text{span}\{1, x\}$; for K_c these two affine modes both have eigenvalue c .

2 The limit inner product and the quotient space

Theorem 2.1 (quadratic form and radical). *For $f \in H^1$,*

$$(f, f)_{1,0} = \int_{-1}^1 |f'|^2 dx - \frac{|f(1) - f(-1)|^2}{2} = \left\| f' - \frac{f(1) - f(-1)}{2} \right\|_{L^2}^2 \geq 0,$$

and $(f, f)_{1,0} = 0$ if and only if $f \in W = \text{span}\{1, x\}$. Consequently W is exactly the radical of $(\cdot, \cdot)_{1,0}$: $(w, f)_{1,0} = 0$ for all $w \in W$, $f \in H^1$.

Proof. Since $f(1) - f(-1) = \int_{-1}^1 f'$, the Cauchy–Schwarz inequality gives $|f(1) - f(-1)|^2 \leq 2\|f'\|_{L^2}^2$, so the first expression is nonnegative. The last identity is the expansion of $\|f' - \alpha\|^2$ with $\alpha = (f(1) - f(-1))/2$ (note $\int \alpha = 2\alpha$ and $\int 1 = 2$). Equality in Cauchy–Schwarz holds iff f' is constant a.e., iff f is affine, iff $f \in W$. For $w \in W$, say $w = \alpha x + \beta$, one has $w' = \alpha$ and $w(1) - w(-1) = 2\alpha$, so $(w, f)_{1,0} = \int \alpha \overline{f'} - \alpha \int \overline{f'} = 0$. $\square$

Theorem 2.2 (isometric isomorphism of the quotient). *The quotient H^1/W equipped with the inner product induced by $(\cdot, \cdot)_{1,0}$ is a Hilbert space, and the map*

$$T: H^1/W \rightarrow L_0^2(-1, 1) := \{g \in L^2(-1, 1) : \int_{-1}^1 g = 0\}, \quad T([f]) := f' - \frac{f(1) - f(-1)}{2},$$

is an isometric isomorphism.

Proof. By Theorem 2.1, $(\cdot, \cdot)_{1,0}$ factors through the quotient and is positive definite there. The map T is well defined and injective: if $T([f]) = 0$ then f' is constant, so f is affine, so $[f] = 0$. It is surjective: for $h \in L_0^2$, $f(x) := \int_0^x h(t) dt$ lies in H^1 with $f' = h$ and $\int h = 0$, so $T([f]) = h$. It is isometric: with $\alpha = (f(1) - f(-1))/2$, $\|T([f])\|_{L^2}^2 = \|f' - \alpha\|^2 = (f, f)_{1,0}$. Since L_0^2 is the kernel of the continuous functional $h \mapsto \int_{-1}^1 h$, it is closed in L^2 . Hence H^1/W is complete. $\square$

3 Complete systems in the quotient

Let P_n be the Legendre polynomials ($P_n(1) = 1$) and set

$$S_0 := 1, \quad S_1 := x, \quad S_n := P_n - P_{n-2} \quad (n \geq 2),$$

the polynomial basis used to construct the Krein–Sobolev polynomials [1, eq. (15)]. Since $\deg S_n = n$ with nonzero leading coefficient, top-down elimination gives the exact decomposition

$$\Pi_N = \text{span}\{1, x\} \oplus \text{span}\{S_2, \dots, S_N\} \quad (N \geq 2).$$

Theorem 3.1 (completeness in the quotient). *(a) The images $\{[S_n] : n \geq 2\}$ span $[\Pi]$ ($= \text{span}\{[p] : p \in \Pi\}$) exactly; (b) $[\Pi]$ is dense in H^1/W ; (c) any orthonormal system $\{Q_n : n \geq 2\}$ obtained from $\{S_n : n \geq 2\}$ by Gram–Schmidt in $(\cdot, \cdot)_{1,0}$ is a complete orthonormal system of H^1/W ; (d) the images of the eigenfunctions, $\{[\cos n\pi x] : n \geq 1\} \cup \{[\sin \mu_n x] : n \geq 1\}$, also form a complete system in H^1/W .*

Proof. (a) is the displayed decomposition (the affine part has image 0). (b) Polynomials are dense in H^1 (standard); the quotient map $H^1 \rightarrow H^1/W$ is continuous because $\|[f]\|_{1,0} \leq \|f'\|_{L^2} \leq \|f\|_{H^1}$, so $[\Pi]$ is dense. (c) follows from (a), (b) and the density of the span in the closure. (d) The actual images under the isometry of Theorem 2.2 are

$$\begin{aligned} T([\cos n\pi x]) &= -n\pi \sin n\pi x, \\ T([\sin \mu_n x]) &= \mu_n \cos \mu_n x - \sin \mu_n. \end{aligned}$$

To prove density in the quotient, fix any $c > 0$. Let $\{e_j\}_{j \geq 0}$ be the L^2 -orthonormal eigenbasis obtained by normalizing the displayed eigenfunctions of K_c , with $e_0 = 1/\sqrt{2}$, $e_1 = \sqrt{3/2}x$, and $K_c e_j = \lambda_j e_j$. Thus $\lambda_0 = \lambda_1 = c$, while each e_j , $j \geq 2$, is a normalized cosine or sine mode listed in (d), with $\lambda_j > c$. For $f \in H^1 = D(K_c^{1/2})$, write $a_j = (f, e_j)_{L^2}$ and set $f_N = \sum_{j=0}^N a_j e_j$, $N \geq 1$. The spectral theorem and (2) give

$$\sum_{j=0}^{\infty} \lambda_j |a_j|^2 = \|f\|_{1,c}^2 < \infty, \quad \|f - f_N\|_{1,c}^2 = \sum_{j>N} \lambda_j |a_j|^2 \longrightarrow 0.$$

Since $(u, u)_{1,c} = (u, u)_{1,0} + c\|u\|_{L^2}^2$ for $u \in H^1$,

$$\|[f - f_N]\|_{1,0}^2 = \|f - f_N\|_{1,c}^2 - c\|f - f_N\|_{L^2}^2 \leq \|f - f_N\|_{1,c}^2 \longrightarrow 0.$$

Both e_0 and e_1 belong to the radical W , so $[e_0] = [e_1] = 0$ (indeed, $T([1]) = 0$ and $T([x]) = 1 - 1 = 0$). Consequently $[f_N] = \sum_{j=2}^N a_j [e_j]$ is a finite linear combination of the classes in (d). As $[f] \in H^1/W$ was arbitrary, this proves their density. $\square$

Lemma 3.2 (explicit orthonormal representatives). *For all $n, m \geq 2$,*

$$S'_n = (2n - 1)P_{n-1}, \quad (S_n, S_m)_{1,0} = 2(2n - 1)\delta_{nm}. \quad (3)$$

Consequently the Gram–Schmidt procedure applied in order to $S_2, S_3, \dots$, with positive leading coefficients, gives the polynomial representatives

$$Q_n := \frac{S_n}{\sqrt{2(2n - 1)}} \quad (n \geq 2). \quad (4)$$

Their classes form a complete orthonormal system of H^1/W .

Proof. Use the Legendre normalization $\int_{-1}^1 P_j P_k = 2\delta_{jk}/(2j+1)$ and leading coefficient $\ell_j = 2^{-j} \binom{2j}{j}$. Since $P_n(1) = P_{n-2}(1) = 1$ and $P_n(-1) = P_{n-2}(-1) = (-1)^n$, both endpoint values of S_n vanish. For any polynomial p of degree at most $n-2$, integration by parts gives

$$\int_{-1}^1 S'_n p = - \int_{-1}^1 (P_n - P_{n-2}) p' = 0.$$

For $n = 2$ this holds because $p' = 0$; for $n \geq 3$ it follows from $\deg p' \leq n-3$ and Legendre orthogonality. Thus S'_n is a multiple of P_{n-1} , and its leading coefficient gives the factor $n\ell_n/\ell_{n-1} = 2n-1$. The boundary term in $(S_n, S_m)_{1,0}$ is zero, so the derivative identity and Legendre orthogonality prove (3). The Gram–Schmidt procedure therefore only divides each S_n by its positive norm, which proves (4). Completeness follows from Theorem 3.1(a)–(c). $\square$

4 Structural stability of the Krein–Sobolev system

The Krein–Sobolev polynomials $\{K_n^{(c)}\}$ are defined by $K_n = a_n S_n + K_{n-2}$, $K_n(1) = 1$, with $a_0 = a_1 = a_2 = a_3 = 1$ and, for $n \geq 2$ [1, eq. (19)],

$$a_{n+2} = a_n \left(1 + \frac{4n^2 - 1}{c} \right) + \frac{2n+1}{2n-3} (a_n - a_{n-2}), \quad (5)$$

and satisfy the orthogonality [1, Theorem 3]

$$(K_m^{(c)}, K_n^{(c)})_{1,c} = \frac{2c}{2n+1} a_n a_{n+2} \delta_{n,m}. \quad (6)$$

Theorem 4.1 (degeneration of the low modes). $K_0^{(c)} = 1$ and $K_1^{(c)} = x$ for all $c > 0$, and $\|K_0^{(c)}\|_{1,c}^2 = 2c$, $\|K_1^{(c)}\|_{1,c}^2 = 2c/3$. Moreover $K_2^{(c)} = P_2$ and $K_3^{(c)} = P_3$ are independent of c , with $\|K_2^{(c)}\|_{1,c}^2 = 6 + \frac{2c}{5} \rightarrow 6$ and $\|K_3^{(c)}\|_{1,c}^2 = 10 + \frac{2c}{7} \rightarrow 10$ as $c \rightarrow 0$.

Proof. The formulas for K_0, K_1 are explicit; their squared norms are $(1,1)_{1,c} = 2c$ and $(x,x)_{1,c} = 2c/3$. From (5) with $a_2 = a_0 = 1$ and $a_3 = a_1 = 1$, $a_4 = 1 + 15/c$ and $a_5 = 1 + 35/c$; the coefficients of S_2, S_0 (resp. S_3, S_1) in K_2 (resp. K_3) are $(1,1)$, so $K_2 = S_2 + S_0 = P_2$ and $K_3 = S_3 + S_1 = P_3$. The norms follow from (6). $\square$

Theorem 4.2 (fixed-index asymptotics and high-mode divergence). For $n = 2m + \epsilon$, where $m \geq 1$ and $\epsilon \in \{0, 1\}$, set

$$r_n := m - 1, \quad C_n := \prod_{j=1}^{m-1} (4(2j + \epsilon)^2 - 1) > 0, \quad (7)$$

with an empty product equal to 1. For every fixed $n \geq 2$, as $c \downarrow 0$,

$$\begin{aligned} a_n(c) &= C_n c^{-r_n} (1 + O_n(c)), \\ \|K_n^{(c)}\|_{1,c}^2 &= 2(2n-1) C_n^2 c^{-2r_n} (1 + O_n(c)). \end{aligned} \quad (8)$$

Here O_n allows constants depending on the fixed index n . In particular, $r_n = (n-2)/2$ for even n and $r_n = (n-3)/2$ for odd n , and the squared norms tend to $+\infty$ for every fixed $n \geq 4$. For $n = 4$ the exact formula is

$$\|K_4^{(c)}\|_{1,c}^2 = \frac{3150}{c^2} + \frac{560}{c} + \frac{80}{3} + \frac{2c}{9}. \quad (9)$$

Proof. Fix $c > 0$. Write $b_n = (2n + 1)/(2n - 3) > 0$ and $\delta_n = a_n - a_{n-2}$ for $n \geq 2$. The recurrence gives

$$\delta_{n+2} = \frac{4n^2 - 1}{c} a_n + b_n \delta_n.$$

Starting with $a_0 = a_2 = 1$, $\delta_2 = 0$ in the even chain and $a_1 = a_3 = 1$, $\delta_3 = 0$ in the odd chain, induction yields $a_n \geq a_{n-2} \geq 1$ and $\delta_n \geq 0$. In particular, division by a_n is legitimate. More precisely, (5) says

$$a_{n+2} = \frac{4n^2 - 1}{c} a_n + R_n, \quad R_n := a_n + b_n(a_n - a_{n-2}), \quad a_n \leq R_n \leq (1 + b_n)a_n. \quad (10)$$

Thus the remainder is $O_n(a_n)$, and

$$c \frac{a_{n+2}}{a_n} = 4n^2 - 1 + c \left(1 + b_n \left(1 - \frac{a_{n-2}}{a_n} \right) \right) = 4n^2 - 1 + O_n(c). \quad (11)$$

For $n = 2, 3$, $r_n = 0$, $C_n = 1$, and the coefficient assertion in (8) is exact. If it holds at n , then (10) gives

$$a_{n+2} = (4n^2 - 1)C_n c^{-(r_n+1)} (1 + O_n(c)).$$

Since $r_{n+2} = r_n + 1$ and $C_{n+2} = (4n^2 - 1)C_n$, this completes the induction on each parity chain, for all fixed n . Substitution in (6) proves the norm assertion, using $2C_n C_{n+2}/(2n + 1) = 2(2n - 1)C_n^2$. When $n \geq 4$, $r_n \geq 1$, so these norms diverge. Finally, $a_4 = 1 + 15/c$ and $a_6 = 1 + 105/c + 945/c^2$ give (9). They also exhibit the failure of an $O(1)$ remainder: $a_6 - (63/c)a_4 = 1 + 42/c$. $\square$

Throughout the next theorem the ordinary Sobolev norm is

$$\|f\|_{H_{\text{Sob}}^1}^2 := \int_{-1}^1 (|f|^2 + |f'|^2) dx.$$

It is distinct from the degenerate seminorm $\|f\|_{1,0}$ and from the c -dependent norm $\|f\|_{1,c}$.

Theorem 4.3 (fixed-index limits of representatives and classes). *Let $\widehat{u}_n^{(c)} := K_n^{(c)}/\|K_n^{(c)}\|_{1,c}$. For every fixed $n \geq 2$, as $c \downarrow 0$,*

$$\widehat{u}_n^{(c)} \longrightarrow L_n \quad \text{in } H_{\text{Sob}}^1, \quad L_n := \begin{cases} P_2/\sqrt{6} = Q_2 + 1/\sqrt{6}, & n = 2, \\ P_3/\sqrt{10} = Q_3 + x/\sqrt{10}, & n = 3, \\ Q_n = S_n/\sqrt{2(2n-1)}, & n \geq 4. \end{cases} \quad (12)$$

In fact $\|\widehat{u}_n^{(c)} - L_n\|_{H_{\text{Sob}}^1} = O_n(c)$. For every fixed $n \geq 2$ the quotient limit is

$$[\widehat{u}_n^{(c)}] \longrightarrow [Q_n] \quad \text{in } (H^1/W, \|\cdot\|_{1,0}), \quad (13)$$

and $\{[Q_n] : n \geq 2\}$ is a complete orthonormal system there. For $n = 0, 1$, $[\widehat{u}_n^{(c)}] = 0$ for every $c > 0$. These assertions are for each fixed index; they make no claim of convergence uniform over n .

Proof. Theorem 4.1 gives

$$\widehat{u}_2^{(c)} = \frac{P_2}{\sqrt{6 + 2c/5}}, \quad \widehat{u}_3^{(c)} = \frac{P_3}{\sqrt{10 + 2c/7}}.$$

The scalar factors differ from $1/\sqrt{6}$ and $1/\sqrt{10}$ by $O(c)$, respectively. This proves the two low-mode limits and rates in the ordinary Sobolev norm. Their affine terms persist in that norm: explicitly,

$$\|L_2 - Q_2\|_{H_{\text{Sob}}^1}^2 = \frac{1}{3}, \quad \|L_3 - Q_3\|_{H_{\text{Sob}}^1}^2 = \frac{4}{15}.$$

Now fix $n = 2m + \epsilon \geq 4$, so $m \geq 2$. Iterating the defining polynomial recurrence is a finite identity:

$$K_n^{(c)} = P_\epsilon + \sum_{j=1}^m a_{2j+\epsilon}(c) S_{2j+\epsilon}, \quad \frac{K_n^{(c)}}{a_n} = S_n + \frac{K_{n-2}^{(c)}}{a_n}. \quad (14)$$

For $1 \leq j \leq m-1$, Theorem 4.2 yields

$$\frac{a_{2j+\epsilon}}{a_n} = \frac{C_{2j+\epsilon}}{C_n} c^{m-j} (1 + O_n(c)), \quad \frac{1}{a_n} = O_n(c^{m-1}).$$

The error constants can be combined because there are only finitely many j for this fixed n . Applying the triangle inequality to (14), and using $m-j \geq 1$ and $m-1 \geq 1$, gives the explicit finite-sum bound

$$\left\| \frac{K_{n-2}^{(c)}}{a_n} \right\|_{H_{\text{Sob}}^1} \leq \frac{\|P_\epsilon\|_{H_{\text{Sob}}^1}}{a_n} + \sum_{j=1}^{m-1} \frac{a_{2j+\epsilon}}{a_n} \|S_{2j+\epsilon}\|_{H_{\text{Sob}}^1} = O_n(c). \quad (15)$$

All coefficient ratios here are positive. By (6) and (11), the positive scalar $\beta_n(c) := \|K_n^{(c)}\|_{1,c}/a_n(c)$ satisfies

$$\beta_n(c)^2 = \frac{2c}{2n+1} \frac{a_{n+2}}{a_n} = 2(2n-1) + O_n(c), \quad \beta_n(c) = \sqrt{2(2n-1)} + O_n(c).$$

Since this limit is positive, its reciprocal also differs from $1/\sqrt{2(2n-1)}$ by $O_n(c)$. Therefore

$$\hat{u}_n^{(c)} = \frac{S_n + K_{n-2}^{(c)}/a_n}{\beta_n(c)} = Q_n + O_n(c) \quad \text{in } H_{\text{Sob}}^1,$$

where the last error denotes a polynomial whose Sobolev norm is $O_n(c)$. This proves the representative assertion for every fixed $n \geq 4$ without taking any infinite-dimensional limit.

Finally the quotient map is continuous, since $\|[f]\|_{1,0} \leq \|f'\|_{L^2} \leq \|f\|_{H_{\text{Sob}}^1}$. For all $n \geq 2$, $L_n - Q_n \in W$, so the representative convergence implies (13), also with an $O_n(c)$ error. Lemma 3.2 proves orthonormality and completeness of the limiting classes. For $n = 0, 1$, the normalized polynomials are still affine and hence have zero quotient class by Theorem 2.1. $\square$

Remark 4.4 (correction of a heuristic). *The natural guess that “ $\|K_n^{(c)}\|_{1,c}$ has a finite limit for all $n \geq 2$ ” is false: it holds only for $n = 2, 3$ (Theorem 4.1); for $n \geq 4$ the norms diverge (Theorem 4.2). The unit-normalized representatives have the limits stated in Theorem 4.3, and their classes converge to $[Q_n]$ for all fixed $n \geq 2$. For $n \geq 4$, dividing by $a_n \rightarrow \infty$ leaves S_n as the leading polynomial, whose endpoint values are zero. The endpoint normalization $K_n(1) = 1$ therefore hides this finite limiting shape. For $n = 2, 3$, the nonzero affine difference between L_n and Q_n disappears only after passage to the quotient.*

Remark 4.5 (verification status and quantifiers). *The fourth-round author repair supplies full analytical proofs of Theorems 4.2 and 4.3; their universal fixed-index assertions do not depend on finite checks. The original proof-sketch status is superseded by the explicit induction and Sobolev remainder estimate above. In particular, the Gram matrix of $S_2, \dots, S_N$ at $c = 0$ is exactly $\text{diag}(6, 10, \dots, 2(2N -$*

1)) for every $N \geq 2$. Its smallest eigenvalue is 6 for every such N ; nonsingularity is not an open obligation. This Gram bound alone is not a uniform-in- n convergence estimate for $\hat{u}_n^{(c)}$. The proofs allow their constants to depend on n and do not address joint limits or infinite sums with indices varying as $c \downarrow 0$. This is an author proof revision, not an independent-acceptance certificate or a claim of full formalization.

Remark 4.6 (spectral interpretation). The two eigenvalues c of K_c (with eigenfunctions $1, x$) converge to the eigenvalue 0 of K_0 , whose eigenspace W becomes the radical of $(\cdot, \cdot)_{1,0}$. On the quotient, the remaining eigenfunctions $\{[\cos n\pi x]\} \cup \{[\sin \mu_n x]\}$ form a complete orthogonal system with squared quotient-norms computable from $(f, f)_{1,0} = (K_0 f, f)_{L^2}$ for $f \in D(K_0)$.

5 Exact checks and their scope

The following identities follow from the analytical proofs and also give finite regression targets. The fourth-round author checks use a separate local symbolic/rational test program; they are supplementary evidence, not proofs of a statement for all indices or independent acceptance. Historical output of `scripts/op08_c0_verify2.py` is not used as a premise for the repaired results.

1. *Radical* (Theorem 2.1). For $f \in \{x^2, S_4, S_7\}$, $(1, f)_{1,0} = (x, f)_{1,0} = 0$ exactly.
2. *Gram of $\{S_2, \dots, S_N\}$* . For every $N \geq 2$, (3) gives determinant $\prod_{n=2}^N 2(2n-1)$ and smallest eigenvalue 6. Thus, for example, the determinant at $N=4$ is $6 \cdot 10 \cdot 14 = 840$.
3. *Norms* (Theorems 4.1, 4.2). The first two squared norms are $2c, 2c/3$, and the next two are $6 + 2c/5, 10 + 2c/7$. Equation (9) gives, at $c = 10^{-3}$, the exact value $3150560026 + 3001/4500$ for $\|K_4\|_{1,c}^2$. Both $n=4$ and $n=5$ have squared-norm growth of order c^{-2} ; for $n=6$ the order is c^{-4} , as dictated by $2r_n$.
4. *Unit-normalized convergence* (Theorem 4.3). The low-mode squared Sobolev discrepancies from Q_2, Q_3 tend to $1/3, 4/15$. For $n=4$, $K_4^{(c)} = a_4 S_4 + S_2 + 1$, so the quotient-orthogonal projection of $[\hat{u}_4^{(c)}]$ onto $\text{span}\{[S_4]\}$ is $(a_4/\|K_4\|_{1,c})[S_4]$. Its coefficient tends to $1/\sqrt{14}$; it is not equal to that limit for $c > 0$. The residual squared quotient norm is exactly $6/\|K_4\|_{1,c}^2 \rightarrow 0$. The ordinary Sobolev convergence also controls the affine remainder, by (15).
5. *Exact spanning* (Theorem 3.1). Top-down elimination with $S_2, \dots, S_N$ leaves an affine remainder for every polynomial of degree at most N , by the nonzero leading coefficients. This is an algebraic proof; finite sample elimination checks only test its implementation.

6 Mathematics involved

Pseudo inner product / radical A Hermitian sesquilinear form $(\cdot, \cdot)$ with $(f, f) \geq 0$; the set of w with $(w, f) = 0$ for all f is the radical. The quotient by the radical carries a genuine inner product.

Cauchy–Schwarz equality case $|\int f' \overline{g'}|^2 \leq \|f'\|^2 \|g'\|^2$ with equality iff f', g' are linearly dependent; this pins down the radical.

Quotient Hilbert space H^1/W with the induced inner product; complete because isometric to the complete space L_0^2 .

Top-down degree elimination $\deg S_n = n$ implies every polynomial decomposes uniquely into an affine part and a combination of $\{S_2, \dots, S_N\}$.

Gram–Schmidt and normalization The divergence of $\|K_n^{(c)}\|$ for $n \geq 4$ shows that the choice of normalization ($K_n(1) = 1$ vs. unit norm) changes the limiting behaviour.

Krein Laplacian The self-adjoint operator $-f''$ with the nonlocal boundary condition $f'(1) = f'(-1) = (f(1) - f(-1))/2$; at $c = 0$ it has a two-dimensional kernel $\text{span}\{1, x\}$.

Spectrum and eigenfunctions $\sigma(K_0) = \{0\} \cup \{(n\pi)^2\} \cup \{\mu_n^2\}$ with $\tan \mu_n = \mu_n$; the transcendental equation is central to the Krein Laplacian spectral analysis.

References

- [1] A. Jones, L. Littlejohn, A. Quintero Roba, *Krein-Sobolev Orthogonal Polynomials II*, Axioms 14 (2025), 115. <https://doi.org/10.3390/axioms14020115>